

Visual Place Recognition: Dataset Landscape Survey

Harshita Jain

Contents

1	Dataset Timeline	3
2	Master Taxonomy	3
3	Outdoor Datasets	5
3.1	Large-Scale Outdoor	5
3.2	Multi-Modal and Range-Based	6
3.3	Urban Retrieval	6
3.4	Compact Baselines	7
4	Indoor Datasets	7
4.1	Static / Relocalization	7
4.2	Repeat-Visit and Change-Aware	8
4.3	Text-Assisted and Repetitive-Structure	9
5	Pedestrian-Centric Datasets	10
6	Adjacent Localization Benchmarks	11
7	Evaluation Protocols and Metrics	11
8	Sensor Modality Matrix	12
9	Gaps and Underrepresented Scenarios	12
9.1	No Unified Indoor Benchmark Protocol	12
9.2	Indoor Long-Term Data Scarcity	12
9.3	Indoor Scale Deficit	13
9.4	Repetitive Structure / Perceptual Aliasing	13
9.5	Text/OCR as a Modality	13
9.6	Outdated Synthetic Pipelines	13

9.7	Privacy / Anonymization	13
9.8	Other Gaps	13

1. Dataset Timeline

Figure 1 provides a chronological overview of when each surveyed dataset was released.

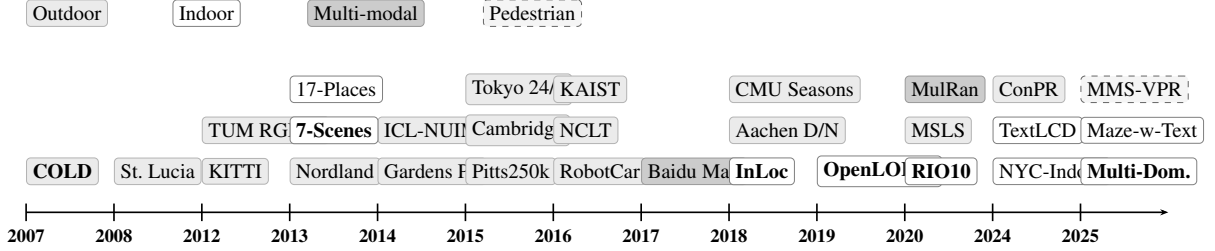


Figure 1: Chronological release timeline of surveyed VPR datasets (2007–2025). Bold labels indicate datasets particularly relevant to indoor VPR benchmarking.

2. Master Taxonomy

Table 1: Master metadata of all surveyed VPR datasets.

Dataset	Year	Scale	Environment / Challenge	Ground Truth
Outdoor — Large-Scale & Long-Term				
Nordland	2013	~3.6 M imgs	728 km train, Norway; sea-sonal	GPS @ 1 Hz
Oxford Robot-Car	2016	~20 M imgs	Oxford urban; weather/season/construction	GPS/INS (RTK 2020)
MSLS	2020	1.6 M imgs	30 cities, 6 continents; 7-yr span	GPS (Mapillary)
NCLT	2016	27 sess., 147 km	Michigan campus; sea-sonal, multi-run	SLAM 6DoF
Outdoor — Multi-Modal & Range-Based				
KAIST All-Day	2016	50 km	KAIST campus; RGB+thermal illumi-nation	GPS/INS
MulRan	2020	4 cities × 3	Korean urban; radar-in-the-loop	LiDAR odom. 6DoF
ConPR	2024	6 sess., multi-yr	Construction site (SNU); structural change	LiDAR SLAM
Outdoor — Urban Retrieval				
Pittsburgh 250k	2016	~254 K imgs	Pittsburgh; wide-baseline (GSV)	Implicit geo-tag
Tokyo 24/7	2015	~76 K imgs	Tokyo; day-vs-night	Implicit geo-tag
Outdoor — Compact Baselines				
St. Lucia	2008	~2.6 K imgs	9.5 km suburban, Australia	GPS loop closure

Dataset	Year	Scale	Environment / Challenge	Ground Truth
Gardens Point	2014	2 × 200 imgs	QUT campus; day/night transition	Frame-index
Indoor — Static / Relocalization				
7-Scenes	2013	7 scenes	Small rooms (Microsoft); static RGB-D	6DoF pose
TUM RGB-D	2012	~39 K fr	TUM office/desk	6DoF (MoCap)
ICL-NUIM	2014	Synthetic	Synthetic room/office	Exact 6DoF
17-Places	2013	406 imgs	Multiple buildings (legacy)	Frame-index
InLoc	2018	9,972 db + 300+ q	Two univ. buildings (WUSTL); furniture	SfM 6DoF (ver- ified)
Indoor — Repeat-Visit & Change-Aware				
RIO10	2020	74 seqs, 10 scenes	Rooms with layout change (3RScan)	RGB-D 6DoF
OpenLORIS- Scene	2019	Multi-sess., 5 types	Service-robot envs; real- world change	MoCap / LiDAR
COLD	2007	Multi-seq, 3 labs	Three EU labs; cross- platform	Odom. + room la- bels
Multi-Domain	2025	89,550 imgs	9 rooms; VPR + anomaly detection	Room/place labels
Indoor — Text-Assisted & Repetitive Structure				
NYC-Indoor- VPR	2024	~36 K imgs	13 NYC scenes, 1 yr; anonymized pedestrians	Semi-auto topo- metric
Maze-with-Text	2025	Panoramic, 5 floors	Multi-floor hallways; scene text spotting	Floor/zone labels
TextLCD	2024	LiDAR + text	Repetitive indoor; text loop closure	LiDAR pose
Baidu Mall	2017	~689 imgs	Indoor mall; dynamic pedestrians	Place labels
Pedestrian-Centric				
MMS-VPR	2025	110 K imgs	Chengdu precinct; pedestrian-only	GPS (WGS84) + text
Adjacent Localization Benchmarks				
KITTI	2012	22 seqs	Karlsruhe driving	RTK GPS/IMU
Cambridge Lmk.	2015	~6–8 K imgs	Cambridge landmarks	SfM 6DoF
Aachen Day– Night	2018	4.5 K db, 922 q	Aachen; day-to-night	SfM 6DoF
CMU Seasons	2018	100 K+ imgs	Pittsburgh suburb; seasonal	SfM/GPS 6DoF

3. Outdoor Datasets

3.1 Large-Scale Outdoor

Nordland (2013) [1]

Authors	Sünderhauf, Neubert, Protzel (Chemnitz UT)
Modalities	RGB, single forward-facing camera
Scale	~3.6 M images; 10 h/season $\times$ 4 seasons, 25 Hz
Scope	728 km Trondheim–Bodø, Norway
Ground truth	GPS @ 1 Hz
Link	https://nrkbeta.no/2013/01/15/nordlandsbanen-minute-by-minute-season-by-season/

Canonical seasonal VPR benchmark. Fixed camera eliminates viewpoint variation, isolating season as the sole variable.

Oxford RobotCar (2016) [2]

Authors	Maddern, Pascoe, Linegar, Newman (Oxford)
Modalities	6 $\times$ RGB (stereo + fisheye), 2D/3D LiDAR, GPS/INS
Scale	~20 M images; 100+ traversals, ~10 km, 1 year
Ground truth	GPS/INS; RTK-corrected 2020
License	CC BY-NC-SA 4.0
Link	https://robotcar-dataset.robots.ox.ac.uk/

Largest camera-centric VPR/SLAM dataset. Weather, illumination, season, construction change on a single repeated route.

MSLS (2020) [3]

Authors	Warburg, Hauberg, López-Antequera et al. (Mapillary/DTU/Zaragoza)
Modalities	RGB, crowd-sourced, compass-derived direction
Scale	1.6 M images; 30 cities, 6 continents, 7-year span
Ground truth	GPS (Mapillary)
Link	https://github.com/mapillary/mapillary_sls

Largest geographically diverse VPR benchmark. Standard for generalizable descriptor training and validation.

NCLT (2016) [4]

Authors	Carlevaris-Bianco, Ushani, Eustice (Michigan)
Modalities	360° RGB, multiple Velodyne LiDARs, IMU, GPS
Scale	27 sessions, 34.9 h, 147 km
Ground truth	6DoF SLAM with covariances
License	ODbL

Long-duration, sensor-rich campus dataset for cross-season retrieval and loop closure.

3.2 Multi-Modal and Range-Based

KAIST All-Day (2016–2018) [5]

Authors	Choi, Kim, Hwang et al. (KAIST)
Modalities	RGB/thermal (co-aligned), stereo RGB, Velodyne 32E, GPS/INS
Scale	50 km, 7.5 h, 25 Hz
Ground truth	GPS/INS
License	CC BY-NC-SA 3.0
Link	https://sites.google.com/view/multispectral/home

Reference RGB+thermal benchmark under extreme illumination change. Single site, no seasonal variation.

MulRan (2020) [6]

Authors	Kim, Park, Cho, Jeong, Kim (KAIST/SNU)
Modalities	3D LiDAR (Ouster), FMCW radar (Navtech), GPS
Scale	4 environments $\times$ 3 sequences
Ground truth	6DoF LiDAR-odometry
License	CC BY-NC-SA 4.0
Link	https://sites.google.com/view/mulran-pr/home

Primary radar/LiDAR-only VPR benchmark. Reverse-loop and lane-change revisits stress rotation invariance.

ConPR (2024) [7]

Authors	Lee, Jung, Kim (SNU)
Modalities	Monocular (30 Hz), 3D LiDAR (Livox), IMU, GPS
Challenge	Structural (terrain/infrastructure) change
Ground truth	LiDAR-SLAM
Link	https://github.com/dongjae0107/ConPR

Rare dataset targeting terrain/infrastructure evolution rather than appearance-only change.

3.3 Urban Retrieval

Pittsburgh 250k (2016) [8]

Authors	Arandjelović, Gronat, Torii, Pajdla, Sivic
Scale	~254 K images from 10,586 GSV panoramas
Ground truth	Implicit geo-location
Link	https://www.di.ens.fr/willow/research/netvlad/

Tokyo 24/7 (2015) [9]

Authors	Torii, Arandjelović, Sivic, Okutomi, Pajdla
Scale	~76 K database; 315 query (day/night pairs)
Ground truth	Implicit geo-location

3.4 Compact Baselines**St. Lucia (2008) [10]**

Authors	Milford, Wyeth (QUT)
Scale	Ref. ~1,549 / query ~1,046; 9.5 km suburban
Ground truth	GPS loop closure

Gardens Point (2014)

Institution	Queensland University of Technology
Scale	2 traverses, 200 images each
Challenge	Day-vs-night, indoor–outdoor transition
Ground truth	Frame-index correspondence

4. Indoor Datasets

Indoor VPR differs structurally from outdoor:

1. **No GPS** — ground truth is 6DoF pose or room labels.
2. **1–2 orders of magnitude smaller** than outdoor benchmarks.
3. **Change drivers:** furniture, lighting, clutter, doors — not season/weather.
4. **Perceptual aliasing:** modern architecture produces identical-looking floors, hallways, stairwells. Geometric verification frequently fails.
5. **Privacy:** high-traffic spaces require pedestrian anonymization.

4.1 Static / Relocalization**7-Scenes (2013) [11]**

Authors	Shotton, Glocker, Zach et al. (Microsoft Research)
Modalities	RGB-D (Kinect)
Scale	7 scenes, 300–800 frames each
Ground truth	6DoF pose

Small, static, single-room. No cross-session change. Often repurposed as indoor VPR sanity check.

TUM RGB-D (2012) [12]

Authors	Sturm, Engelhard, Endres, Burgard, Cremers (TUM)
Modalities	RGB-D (Kinect)
Scale	~39 K frames
Ground truth	6DoF (MoCap)
Link	https://cvg.cit.tum.de/data/datasets/rgbd-dataset

Standard SLAM/odometry benchmark. Larger than 7-Scenes but single-session.

ICL-NUIM (2014) [13]

Authors	Handa, Whelan, McDonald, Davison (Imperial College)
Modalities	Synthetic RGB-D
Ground truth	Exact synthetic 6DoF

Purely synthetic; no real sensor noise or scene change. Outdated — see §9 on modern synthetic needs.

17-Places (~2013)

Scale	406 images, multiple buildings
Ground truth	Frame-index correspondence

Legacy benchmark.

InLoc (2018) [14]

Authors	Taira, Okutomi, Sattler et al. (Tokyo Tech / ETH / CIIRC)
Modalities	RGB-D database (WUSTL scans), RGB phone queries
Scale	9,972 database + 300+ queries across two buildings
Ground truth	SfM 6DoF, manually verified
License	ODbL
Link	https://github.com/HajimeTaira/InLoc_dataset

Standard large-scale indoor localization benchmark. Temporal gap between database and queries introduces natural furniture/illumination change.

4.2 Repeat-Visit and Change-Aware**RIO10 (2020) [15, 16]**

Authors	Wald, Sattler, Golodetz, Cavallari, Tombari (TUM / CTU / Niantic / Google)
Modalities	RGB-D sequences
Scale	74 sequences, 10 scenes
Challenge	Furniture moved/replaced, clutter, illumination over time
Ground truth	6DoF pose
Link	https://waldjohannau.github.io/RIO10/

Most direct indoor counterpart to outdoor cross-season benchmarks. Isolates scene change over time as the central difficulty.

OpenLORIS-Scene (2019) [17]

Authors	Shi, Li, Zhao et al. (Intel Labs China + partners)
Modalities	RGB, depth, IMU, wheel odometry
Scale	Multiple scene types (home, office, corridor, cafe, market), multiple visits
Challenge	Lifelong SLAM with out-of-sight scene changes
Ground truth	MoCap / LiDAR trajectory
Link	https://lifelong-robotic-vision.github.io/dataset/scene.html

Service-robot motivated. Same locations across sessions with everyday changes.

COLD (2007–2009) [18]

Authors	Pronobis, Caputo et al. (Ljubljana / Freiburg / DFKI Saarbrücken)
Modalities	Perspective + omnidirectional cameras, laser scans, odometry
Scale	Multiple sequences per lab, varying illumination/time-of-day
Challenge	Cross-lab, cross-platform room recognition
Ground truth	Odometry + laser + room labels
Link	https://www.coldb.org/

Oldest dedicated indoor place-recognition benchmark. Three independent labs, different robots — useful for cross-domain generalization.

Multi-Domain Indoor VPR/AD (2025) [19]

Authors	Woźniak, Krzeszowski, Kwolek (Rzeszów UT / AGH Kraków)
Modalities	RGB (manual + robot-mounted)
Scale	89,550 images, 9 rooms
Challenge	Lighting, viewpoint, human activity; paired VPR + anomaly detection
Ground truth	Room/place labels
Reference	<i>Sci. Data</i> 12, 817 (2025)

Most recent purpose-built indoor VPR dataset. Largest indoor image count surveyed (89,550). Pairs VPR with anomaly detection.

4.3 Text-Assisted and Repetitive-Structure

Scene text (room numbers, signs, floor indicators) is emerging as a disambiguation modality for perceptual aliasing in architecturally repetitive indoor environments.

NYC-Indoor-VPR (2024) [20]

Modalities	RGB (pedestrians masked with white pixels)
Scale	~36,000 images, 13 crowded scenes, 1 year
Challenge	Dense pedestrian obstruction; crowded indoor
Ground truth	Semi-automatic topometric annotation
Privacy	Pedestrians anonymized

Critical for VPR under pedestrian obstruction. Anonymization protocol is a methodological contribution.

Maze-with-Text (2025) [21]

Modalities	Panoramic (cropped to 90° FOV)
Scale	Multi-floor building, 5 floors
Challenge	Repetitive hallway structure; scene text spotting (STS)
Ground truth	Floor/zone labels

Targets the multi-floor aliasing problem where geometric verification fails. Demonstrates STS as a viable disambiguation modality.

TextLCD (2024) [22]

Modalities	LiDAR + scene text
Challenge	Loop closure in structurally repetitive environments
Ground truth	LiDAR pose

Multi-modal (LiDAR + text) loop closure. Text cues complement range sensing in aliased environments.

Baidu Mall (2017) [23]

Modalities	RGB
Scale	~689 images, indoor commercial mall
Challenge	Dynamic pedestrian obstruction
Ground truth	Place labels

Older but heavily cited indoor VPR dataset with dynamic pedestrians. Historical baseline for crowded commercial scenes.

5. Pedestrian-Centric Datasets

MMS-VPR (2025) [24]

Authors	Ou, Ren, Sun et al. (Auckland / HIT Shenzhen)
Modalities	RGB, video clips, GPS, timestamps, textual metadata
Scale	110,529 images; 2,527 clips; 208 locations
Scope	Chengdu Taikoo Li, ~70,800 m ² pedestrian precinct
Ground truth	GPS (WGS84) + JSON metadata
License	CC BY-NC-SA 4.0
Link	https://huggingface.co/datasets/Yiwei-Ou/MMS-VPR

First large-scale pedestrian-only VPR dataset. Non-Western urban context; addresses historical bias toward vehicle-mounted imagery.

6. Adjacent Localization Benchmarks

Metric-localization/SLAM benchmarks routinely used as retrieval baselines in VPR literature.

Table 2: Adjacent benchmarks frequently cited in VPR work.

Dataset	Year	Scope	Reference
KITTI	2012	Karlsruhe driving, RTK-GPS	Geiger et al. [25]
Cambridge Lmk.	2015	6 outdoor scenes, SfM pose	Kendall et al. [26]
Aachen Day–Night	2018	Day-to-night, SfM GT	Sattler et al. [27]
CMU Seasons	2018	Suburban driving, seasonal	Badino et al. [28]

7. Evaluation Protocols and Metrics

Table 3: Evaluation protocols for VPR.

Protocol	Description	Metrics	Typical Datasets
Image Retrieval	Query vs. reference database; positives by geo/pose threshold	Recall@N, P-R, F1	Pitts250k, Tokyo24/7, MSLS, MMS-VPR
Sequential	Sliding-window sequence matching (SeqSLAM-style)	Seq. P/R, AUC	Nordland, RobotCar, NCLT
Metric Pose	Estimate 6DoF camera pose from visual data + map	Pose error (m/°), % < thresh.	RobotCar, Aachen, InLoc, RIO10, 7-Scenes
Cross-Condition	Robustness to season/lighting change	Recall@1 per condition	Nordland, MSLS, RobotCar, Tokyo24/7
Multi-Modal	LiDAR/radar/thermal, possibly fused with RGB	Recall@N or pose error	MulRan, ConPR, KAIST
Indoor Repeat-Visit	Same scene re-visited; robustness to furniture/clutter/lighting	Reloc. acc., room-level acc.	RIO10, OpenLORIS, COLD, Multi-Domain
Text-Assisted	Scene text (OCR/STS) resolves perceptual aliasing	Recall@N, loop closure prec.	Maze-w-Text, TextLCD, NYC-Indoor
Standardized Splits	VPR-Bench [29]: normalized dataset splits and harnesses for fair comparison	Recall@N, EP, AUC	Corridor, Living Room splits

8. Sensor Modality Matrix

Table 4: Sensor modalities across surveyed datasets. **Text/OCR** tracked as emerging modality.

Dataset	RGB	Depth	Thermal	LiDAR	Radar	GPS/IMU	Text
Nordland	✓	✗	✗	✗	✗	✓	✗
Oxford RobotCar	✓	✗	✗	✓	✗	✓	✗
MSLS	✓	✗	✗	✗	✗	✓	✗
NCLT	✓	✗	✗	✓	✗	✓	✗
KAIST All-Day	✓	✗	✓	✓	✗	✓	✗
MuRan	✗	✗	✗	✓	✓	✓	✗
ConPR	✓	✗	✗	✓	✗	✓	✗
Pittsburgh 250k	✓	✗	✗	✗	✗	✗	✗
Tokyo 24/7	✓	✗	✗	✗	✗	✗	✗
St. Lucia	✓	✗	✗	✗	✗	✓	✗
Gardens Point	✓	✗	✗	✗	✗	✗	✗
7-Scenes	✓	✓	✗	✗	✗	✗	✗
TUM RGB-D	✓	✓	✗	✗	✗	✓	✗
ICL-NUIM	✓	✓	✗	✗	✗	✗	✗
InLoc	✓	✓	✗	✗	✗	✗	✗
RIO10	✓	✓	✗	✗	✗	✗	✗
OpenLORIS-Scene	✓	✓	✗	✗	✗	✓	✗
COLD	✓	✗	✗	✗	✗	✓	✗
Multi-Domain	✓	✗	✗	✗	✗	✗	✗
NYC-Indoor-VPR	✓	✗	✗	✗	✗	✗	✗
Maze-with-Text	✓	✗	✗	✗	✗	✗	✓
TextLCD	✗	✗	✗	✓	✗	✗	✓
Baidu Mall	✓	✗	✗	✗	✗	✗	✗
MMS-VPR	✓	✗	✗	✗	✗	✓	✓
KITTI	✓	✗	✗	✓	✗	✓	✗

9. Gaps and Underrepresented Scenarios

9.1 No Unified Indoor Benchmark Protocol

Outdoor VPR has converged on Recall@N against MSLS/Pittsburgh splits. Indoor datasets use pose error (InLoc, RIO10), robustness splits (OpenLORIS), and room-label accuracy (COLD, Multi-Domain) independently. VPR-Bench [29] provides standardized splits but adoption is uneven. A cross-dataset benchmarking pass is a concrete open contribution.

9.2 Indoor Long-Term Data Scarcity

Only RIO10, OpenLORIS, COLD, NYC-Indoor-VPR, and Multi-Domain explicitly model repeat visits over time — a fraction of what exists for outdoor seasonal change.

9.3 Indoor Scale Deficit

The largest indoor-specific set surveyed (89,550 images) is below Pittsburgh 250k and far below MSLS or RobotCar.

9.4 Repetitive Structure / Perceptual Aliasing

Hallways on the 2nd floor are visually identical to the 5th. Geometric verification fails. Maze-with-Text and TextLCD target this, but there is no standard benchmark or metric for measuring aliasing robustness.

9.5 Text/OCR as a Modality

Scene text (room numbers, signs) is becoming a standard strategy against repetitive structures. Most modality surveys omit it. Table 4 tracks it.

9.6 Outdated Synthetic Pipelines

ICL-NUIM (2014) is the only synthetic indoor dataset surveyed — static and outdated. Modern scalable synthetic generation (e.g., injecting dynamic anomalies into COCO-style annotations) is unaddressed.

9.7 Privacy / Anonymization

Indoor VPR in high-traffic spaces faces data-privacy constraints. NYC-Indoor-VPR uses white-pixel masking; most other datasets do not document anonymization status. Future datasets should state anonymization protocols explicitly.

9.8 Other Gaps

- **Radar/LiDAR:** Beyond MulRan, no multi-city radar VPR benchmark exists.
- **Aerial/UAV:** Essentially unaddressed in the current landscape.
- **Structural change:** ConPR (outdoor) and RIO10/OpenLORIS (indoor) are near-sole examples; minority use case vs. lighting/season benchmarks.

References

- [1] Sünderhauf, N., Neubert, P., Protzel, P. (2013). *Are we there yet? Challenging SeqSLAM on a 3000 km journey across all four seasons*. Proc. Workshop on Long-Term Autonomy, IEEE ICRA.
- [2] Maddern, W., Pascoe, G., Linegar, C., Newman, P. (2017). *1 Year, 1000 km: The Oxford RobotCar Dataset*. IJRR 36(1).
- [3] Warburg, F., Hauberg, S., López-Antequera, M. et al. (2020). *Mapillary Street-Level Sequences: A Dataset for Lifelong Place Recognition*. CVPR 2020.
- [4] Carlevaris-Bianco, N., Ushani, A.K., Eustice, R.M. (2016). *University of Michigan North Campus Long-Term Vision and LIDAR Dataset*. IJRR 35(9).
- [5] Choi, Y. et al. (2018). *KAIST Multi-Spectral Day/Night Data Set for Autonomous and Assisted Driving*. IEEE T-ITS 19(3).

- [6] Kim, G., Park, Y.S., Cho, Y., Jeong, J., Kim, A. (2020). *MulRan: Multimodal Range Dataset for Urban Place Recognition*. ICRA 2020.
- [7] Lee, D., Jung, M., Kim, A. (2024). *ConPR: Ongoing Construction Site Dataset for Place Recognition*. arXiv:2407.03684.
- [8] Arandjelović, R., Gronat, P., Torii, A., Pajdla, T., Sivic, J. (2016). *NetVLAD: CNN Architecture for Weakly Supervised Place Recognition*. CVPR 2016.
- [9] Torii, A., Arandjelović, R., Sivic, J., Okutomi, M., Pajdla, T. (2018). *24/7 Place Recognition by View Synthesis*. IEEE TPAMI 40(2).
- [10] Milford, M., Wyeth, G.F. (2008). *Mapping a Suburb With a Single Camera Using a Biologically Inspired SLAM System*. IEEE T-RO 24(5).
- [11] Shotton, J. et al. (2013). *Scene Coordinate Regression Forests for Camera Relocalization in RGB-D Images*. CVPR 2013.
- [12] Sturm, J., Engelhard, N., Endres, F., Burgard, W., Cremers, D. (2012). *A Benchmark for the Evaluation of RGB-D SLAM Systems*. IROS 2012.
- [13] Handa, A., Whelan, T., McDonald, J., Davison, A.J. (2014). *A Benchmark for RGB-D Visual Odometry, 3D Reconstruction and SLAM*. ICRA 2014.
- [14] Taira, H. et al. (2018). *InLoc: Indoor Visual Localization with Dense Matching and View Synthesis*. CVPR 2018.
- [15] Wald, J., Sattler, T., Golodetz, S., Cavallari, T., Tombari, F. (2020). *Beyond Controlled Environments: 3D Camera Re-Localization in Changing Indoor Scenes*. ECCV 2020.
- [16] Wald, J. et al. (2019). *RIO: 3D Object Instance Re-Localization in Changing Indoor Environments*. ICCV 2019.
- [17] Shi, X. et al. (2019). *Are We Ready for Service Robots? The OpenLORIS-Scene Datasets for Lifelong SLAM*. arXiv:1911.05603.
- [18] Pronobis, A., Caputo, B. (2009). *COLD: The CoSy Localization Database*. IJRR.
- [19] Woźniak, P., Krzeszowski, T., Kwolek, B. (2025). *Multi-Domain Indoor Dataset for VPR and Anomaly Detection by Mobile Robots*. Sci. Data 12, 817.
- [20] Díaz-Romero, D.A., Ochoa-Ruiz, G. et al. (2024). *NYC-Indoor-VPR: A Long-Term Indoor Visual Place Recognition Dataset*.
- [21] (2025). *Maze-with-Text: VPR Benchmark for Multi-Floor Repetitive Structures with Scene Text*.
- [22] (2024). *TextLCD: Text-Assisted Loop Closure Detection in Structurally Repetitive Environments*.
- [23] Sun, B., Song, S. et al. (2017). *Indoor Visual Place Recognition via Dense Matching*.
- [24] Ou, Y. et al. (2025). *MMS-VPR: Multimodal Street-Level Visual Place Recognition Dataset and Benchmark*. arXiv:2505.12254.
- [25] Geiger, A., Lenz, P., Urtasun, R. (2012). *Are We Ready for Autonomous Driving? The KITTI Vision Benchmark Suite*. CVPR 2012.
- [26] Kendall, A., Grimes, M., Cipolla, R. (2015). *PoseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalization*. ICCV 2015.

- [27] Sattler, T. et al. (2018). *Benchmarking 6DOF Outdoor Visual Localization in Changing Conditions*. CVPR 2018.
- [28] Badino, H., Huber, D., Kanade, T. (2011). *The CMU Visual Localization Data Set*.
- [29] Zaffar, M. et al. (2020). *VPR-Bench: An Open-Source Visual Place Recognition Evaluation Framework*. arXiv:2005.08135.